Energy 211 (2020) 118710

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Synergistic use of carbon dioxide in catalytic pyrolysis of *chlorella vulgaris* over Ni and Co catalysts

Sungyup Jung [a], Jung-Hun Kim [a], Young Jae Jeon [b], Young-Kwon Park [c], Eilhann E. Kwon [a, *]

[a] Department of Environment and Energy, Sejong University, Seoul, 05006, Republic of Korea
[b] Department of Microbiology, Pukyong National University, Busan, 48513, Republic of Korea
[c] School of Environmental Engineering, University of Seoul, Seoul, 02504, Republic of Korea

ARTICLE INFO

Article history:
Received 17 June 2020
Received in revised form
29 July 2020
Accepted 22 August 2020
Available online 27 August 2020

Keywords:
Microalgae
Waste valorization
Carbon dioxide
Catalytic pyrolysis
Biochar fabrication
Syngas

ABSTRACT

Recovering energy from microalgal biomass via pyrolysis offers an active precautious measure for global warming due to its superior carbon fixation capability. Hence, catalytic pyrolysis of *chlorella vulgaris* (*C. vulgaris*) over Co and Ni catalysts was done to recover energy as a form of syngas. To offer a more sustainable measure for syngas formation, possible use of CO_2 as a reaction feedstock was examined. As such, functional effectiveness of CO_2 on *C. vulgaris* pyrolysis was mainly scrutinized. From non-catalytic pyrolysis of *C. vulgaris*, CO_2 turned into CO by its reduction and simultaneous oxidation of volatile pyrolysates produced from *C. vulgaris* pyrolysis. Such the effectiveness of CO_2 also resulted in the more gaseous pyrogenic products (syngas and C_{1-2} hydrocarbons), but its effectiveness was restrictively initiated at $\geq$ 510 °C due to retarded reaction kinetics. To accelerate the reaction kinetics governing the formation of gaseous pyrogenic products, earth abundant and non-toxic metal Co and Ni catalysts were adopted for catalytic pyrolysis. The significant improvement of syngas production was achieved from Ni (5.6 times) and Co (2.6 times) catalysts, respectively, in reference to non-catalytic pyrolysis under the CO_2 environment.

1. Introduction

Anthropogenic CO_2 emissions derived from the diverse use of fossil resources to fulfil global energy demands have been perceived as significant because CO_2 emissions are considered as one of primary contributors resulting in global warming [1]. Such the CO_2 influxes to the atmosphere are far beyond the nature's spontaneous capacity to retain the atmospheric CO_2 concentration [2]. Thus, there have been unprecedented efforts to retard and/or prevent global warming. For example, carbon capture and storage (CCS) such as CO_2 separation, adsorption, delivery and storage have been proposed/developed for the last couple of decades [3]. For example, high cost of CCS from CO_2 adsorption and separation steps has been considered one of the main constraints for the its practical applications [4].

Among the diverse CO_2 sorptive materials, carbon-rich material (biochar) derived from biomass pyrolysis, has gained a considerable attention due to its high CO_2 adsorption capacity and low production cost [5,6]. Indeed, pyrolysis of biomass offers a facile valorization measure by means of energy recovery from it (syngas and biocrude), while producing biochar as by-product [4,7]. Accordingly, the fundamental researches on biochar in line with the resulting gaseous and liquid pyrolysates are required in the overall process levels to maximize the utilization of biomass-derived useful products.

Considering that pyrolysis is one of the conventional fuel processing techniques by means of thermally induced rearrangement of carbons in a raw feedstock into three phase pyrogenic products (char, oil, and gas) [4,8], it is highly desirable to understand how carbons in a raw feedstock are distributed to each pyrogenic product [9,10] in that these research efforts offer a tactical means for distribution of three phase pyrolysates. Indeed, pyrolysis of biomass requires high thermal energy (400-1000 °C) under an oxygen-free environment [11,12]. As such, seeking an environmentally friendly process is of great importance. More preferably, it is environmentally beneficial to use greenhouse gas, such as CO_2, as a raw feedstock may fortifying the technical viability because such

* Corresponding author.
E-mail address: ekwon74@sejong.ac.kr (E.E. Kwon).

https://doi.org/10.1016/j.energy.2020.118710

the practice using CO_2 offers an innovative way to realize CO_2-to-fuel/chemical [13].

Indeed, huge arable land and enormous quantity of clean water is required to obtain terrestrial biomass [14]. On the contrary to terrestrial biomass, microalgae is relatively less vulnerable to such the restraints [15]. Also, microalgal biomass has much better photosynthesis efficiency than other types of biomass [16], making it a strong candidate for CO_2 fixation. Thus, it is supposed that more carbon could be assimilated within the microalgal biomass in comparison to other ones [17]. 1 tons of microalgae can integrate 1.83 tons of CO_2 for its growth through the photosynthesis [18].

Based on these rationales, pyrolysis of *C. vulgaris* was conducted under the CO_2 environment as a case study. The functional effectiveness of CO_2 on *C. vulgaris* pyrolysis and resulting solid, liquid, and gaseous pyrolysates were qualitatively/quantitatively analyzed to examine the feasibility of *C. vulgaris* and CO_2 for use as the raw feedstocks of syngas and biochar. To improve the production rate of syngas, catalytic pyrolysis was conducted over earth abundant and non-toxic catalysts (Co and Ni).

2. Methods and materials

2.1. Materials and chemical reagents

C. vulgaris sample (freeze dried) was obtained from National Academy of Agricultural Sciences Korea. Prior to pyrolysis study, it was pulverized to have a powdered sample ($\leq$0.1 mm) and kept in a 90 °C convection oven at least 2 d. Dichloromethane (>99.9%), nitric acid (>69%), and silica support (high purity grade, pore size: 60 Å) were acquired from Sigma Aldrich (USA). Nickel(II) nitrate hexahydrate (>97%) and cobalt(II) nitrate hexahydrate (>98%) were purchased from Daejung (Korea). UHP grade N_2 and CO_2 gases, and 20 vol% H_2 in N_2 balance were purchased from Green Gas (Korea).

2.2. Thermolytic profiles of C. vulgaris under N_2/CO_2 atmospheres

The thermal degradation of *C. vulgaris* was scrutinized as a thermolytic temperature using a thermo-gravimetric analysis (TGA: Netzsch STA 449) unit under 3 atm (N_2/CO_2/air). 10.00 mg ($\pm$0.05) of powdered *C. vulgaris* sample was placed in the TGA instrument for thermal degradation. Heating ramp rate used for thermolysis between 35 and 900 °C was 10 °C min^{-1}. Prior to TGA test, a blank TGA run (without *C. vulgaris* sample) was done under each atmosphere, and the results were used as reference plots to subtract buoyancy effects. The proximate analysis of the microalgal biomass was performed comparing TGA results under the two environments (N_2 and Air): volatile matters (73.1 wt%), fixed carbon (15.7 wt%), ash (9.3 wt%), and moisture (1.9 wt%).

2.3. Synthesis of Ni and Co-based catalysts

5 wt% Ni/SiO_2 and 5 wt% Co/SiO_2 catalysts were prepared by the wetness incipient impregnation method [19]. Briefly, each nickel precursor and cobalt precursor was dissolved in the deionized water, and the fully mixed precursor solutions were slowly impregnated onto the silica support. Impregnated samples were dried (100 °C) overnight. Nickel sample was calcined at 600 °C for 5 h under the air condition, followed by the reduction of it under the 20% H_2 (N_2 balance) at 510 °C with a heating of 2 °C min^{-1}. For the preparation of Co/SiO_2 catalyst, calcination was done at 550 °C, and the reduction was performed at 550 °C with a 2 °C min^{-1} of heating rate under the same atmosphere with procedure for Ni/SiO_2 catalyst preparation. ICP-OES instrument (PerkinElmer Optima 5300 DV) was used to determine the Ni (4.9 ± 0.1 wt%) and Co (5.0 ± 0.1 wt%) loading on the silica.

2.4. C. vulgaris pyrolysis and product analysis

C. vulgaris was pyrolyzed using a laboratory made tubular reactor system, equipped with external heating furnaces. Detailed experimental setup was described elsewhere [20]. Briefly, quartz tubes used in the reactor setup were 60 and 90 cm long for each one-stage and two-stage pyrolysis, respectively. For the one-stage setup of *C. vulgaris* pyrolysis, 1.00 g ($\pm$0.01) of sample was placed in center of the quartz tube, which was operated as the first heating zone from 35 to 720 °C with 10 °C min^{-1} of heating rate. An additional heating unit was used to provide thermal energy (isothermally ran) right next to the first heating unit for two-stage pyrolysis. In addition, Ni/SiO_2 or Co/SiO_2 catalyst (1.00 ± 0.01 g) was placed in the middle of the second heating zone for catalytic pyrolysis of volatile matters (VMs) liberated from the first heating zone. Purge gas flow rate was constant at 200 mL min^{-1}. The tubular reactor was directly connected to a micro-GC (3000A, Inficon) for analysis of gaseous effluents evolved from *C. vulgaris* thermolysis. Condensable pyrolysates were collected using a solvent trap (dichloromethane, 0 °C), which was connected to the tubular reactor, and then the samples were analyzed by GC-TOF/MS (7850B, Agilent). To examine the solid biochar produced from *C. vulgaris* pyrolysis, FE-SEM/EDX (Hitachi S-4700, Japan) was utilized. Inorganic components involved in *C. vulgaris* was analyzed using an ICP-OES analysis: Ti (0.6 mg g^{-1}), Mg (3.3 mg g^{-1}), and K (5.1 mg g^{-1}).

3. Results and discussion

3.1. Thermolytic profiles of C. vulgaris under the three atmospheres

The thermolytic behaviour of *C. vulgaris* was characterized thermo-gravimetrically prior to pyrolysis study. As shown in Fig. 1, mass decays of *C. vulgaris* from 3 atm (N_2/CO_2/air) were scrutinized as a function of thermolysis temperature. Also, its differential thermograms (DTGs) from the three environments were presented. Fig. 1(a) demonstrates that the thermolytic behaviours of *C. vulgaris* at $\leq$ 820 °C from the N_2/CO_2 environments are same. Since the TGA apparatus is capable of differentiating the sub-microgram mass difference in the sample [21], any heterogeneous reactions likely result in the difference in the thermolytic patterns. Such the same thermal degradation patterns of *C. vulgaris* at $\leq$ 820 °C from both atmospheres are indicative of no effectiveness of CO_2 on heterogeneous reactions with *C. vulgaris* considering that N_2 is regarded as an inert gas through the thermo-chemical reaction. The residual mass according to the temperature changes in Fig. 1(a) offers that the moisture and volatile matters (VMs) contents of *C. vulgaris* are 1.9 and 74.1 wt%. Specifically, the major mass change is initiated at $\geq$ 250 °C from both environments, which indicates that the generation of gaseous effluents is being initiated at $\geq$ 250 °C. Also, the mass change of *C. vulgaris* at $\geq$ 420 °C starts to be delayed. Such the phenomenon strongly offers that the biochar formation derived from *C. vulgaris* pyrolysis is actively initiated at $\geq$ 420 °C. Considering that the biochar formation results from carbonization, its formation can be interpreted as aromaticity increase [22,23]. Thus, the biochar formation is being ripened at $\geq$ 420 °C through the dehydrogenation mechanisms [9]. Nonetheless, the TGA tests of *C. vulgaris* cannot offer a key evidence of dehydrogenation in line with the formation of biochar. It is noted that the TGA test of *C. vulgaris* only offers the experimental data depicting the mass change of it according to the given temperatures controlled by the external heating source.

Only from the CO_2 environment, the additional mass change is achieved at $\leq$ 820 °C. This is likely ascribed to the gasification reaction of CO_2 with solid carbon ($C(s) + CO_2 \rightleftharpoons 2CO$), which is called

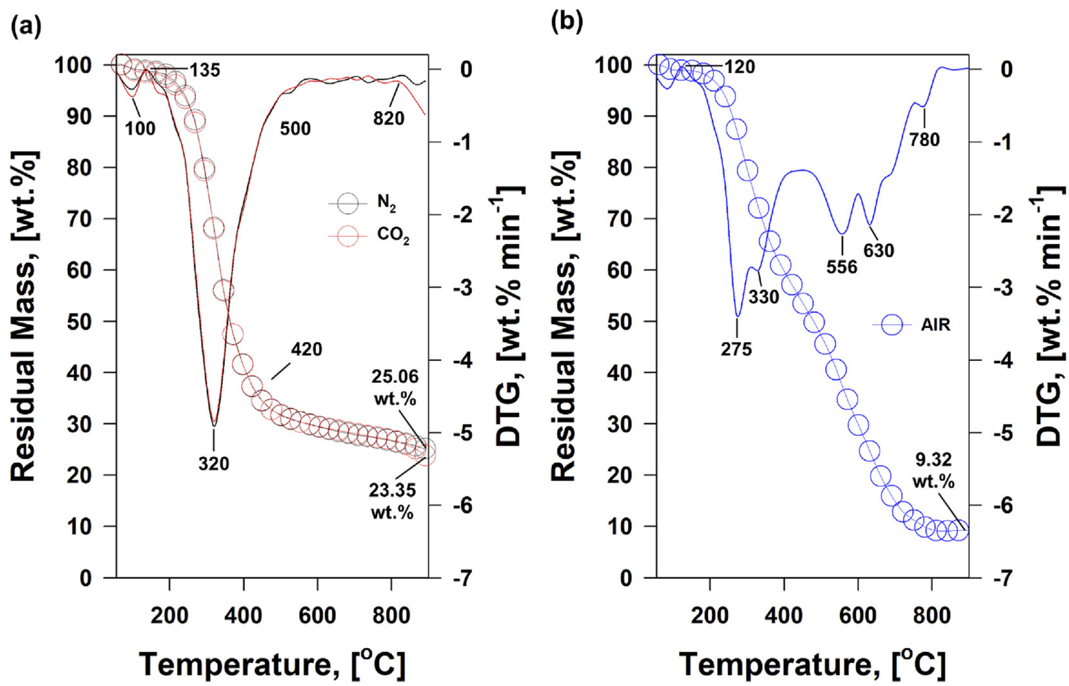

Fig. 1. Mass decay and differential thermogram (DTG) curves of *C. vulgaris* in different atmospheres.

the Boudouard reaction (BR) that is thermodynamically spontaneous at ≥ 710 °C [24]. However, the reaction kinetics of the BR highly depends on the surface morphology of carbon substrates [25] due to different mass transfer of CO_2 through their surface morphology and crystalline structures [25]. Accordingly, the initiation of the BR at ≥ 820 °C indicates the slow reaction kinetics. Also, due to its retarded reaction kinetics, the final residual mass of *C. vulgaris* even at 900 °C was 23.4 wt% from the CO_2 environment, while the final residual mass under the N_2 environment was 25.06 wt%. Such the observation demonstrates that the carbon rearrangement of *C. vulgaris* into CO by the BR is not done. As indicated in Fig. 1(b), the air atmosphere shows that the residual mass (*i.e.*, ash content) of *C. vulgaris* was 9.3 wt%, and such the observation implies that 14.1 wt% of carbon is preserved in the biochar sample even after the initiation of the BR due to its retarded reaction kinetics.

3.2. Pyrolysis of C. vulgaris under the CO_2 and N_2 environments

Although the TGA tests offered the important thermolytic features of *C. vulgaris* from the N_2/CO_2 conditions, the data interpretations from them were only limited to the physical aspects characterizing the thermal degradation of *C. vulgaris*. For an in-depth study, the pyrolysis tests of *C. vulgaris* under the 2 atm were conducted. In addition, major gaseous pyrolysates from *C. vulgaris* pyrolysis were quantified as a function of temperature in Fig. 2(a–e). Moreover, the overall mass balances of three phase pyrogenic products (gas, biocrude, and biochar) were presented in Fig. 2(f). It is noted that the pyrolysis tests of *C. vulgaris* from N_2/CO_2 environments were done at a constant heating rate of 10 °C min^{-1} between 35 and 720 °C. This temperature range is chosen to avoid any effectiveness of CO_2 through the BR.

As Fig. 2(a–e) well demonstrated, the evolution trends of the quantified five gaseous products were nearly identical at ≤ 540 °C under the 2 atm. The evolution of H_2 was initiated at ≥ 420 °C, which is likely ascribed to the dehydrogenation. Such the observations agree with the TGA experimental results. We claimed that the formation of biochar was expected to be initiated at ≥ 420 °C by means of the dehydrogenation. The evolution of H_2 increase as the experimental temperature increase. Such the evolution trend offers that the reaction kinetics of the dehydrogenation is enhanced by the pyrolysis temperature, which is in good agreement with the typical thermolysis of a carbonaceous material [21,26]. The formations of CO and hydrocarbon species are also likely derived from bond scissions of polymeric backbone of *C. vulgaris* [27]. In reference to the evolution temperature of H_2, the evolution temperatures of CO and hydrocarbon products are lower. These the evolution trends are explained by the bonding energy differences. It is noted that bonding energies of C–H, C–O, and C–C are 413, 358, and 347 kJ mol^{-1}, respectively [28,29].

One of the most interesting observation in Fig. 2(c) is that the evolution of CO only from the CO_2 environment is enhanced at ≥ 540 °C. Such the noticeable generation of CO at ≥ 540 °C under the CO_2 environment cannot be recognized without an additional oxygen source. Thus, most oxygen source from *C. vulgaris* is likely depleted from 270 to 510 °C. In detail, from the N_2/CO_2 environments, the generation of CO reached to the maximum at 360 °C and started to drop at ≥ 360 °C. Accordingly, such the CO evolution trends are interpreted as a source depletion in a batch-type reaction setup. Given that the instrument setup, the only oxygen source for CO production is CO_2. From the TGA tests from two different atmospheric conditions, the identical thermolytic profiles of *C. vulgaris* were observed at ≤ 820 °C. Furthermore, the initiation of the BR was shown at ≥ 820 °C from the TGA tests. Hence, the more CO generation from the CO_2 environment is the outcome of the homogeneous gas phase reaction of CO_2 with VMs derived from the thermolysis of *C. vulgaris* for a certain reason. As such, we presumed that reduction of CO_2 into CO and oxidation of VMs through the gas phase reaction occurred simultaneously. Also, Fig. 2(a) shows the simultaneous generation of H_2 and its suppression at the temperature range (≥630 °C)exhibiting the enhanced CO generation from the CO_2 condition. Such the effectiveness of CO_2 signifies that CO_2 provides a new way to manipulate the ratio of carbon and hydrogen by means of consuming carbons in all pyrolysates (except for

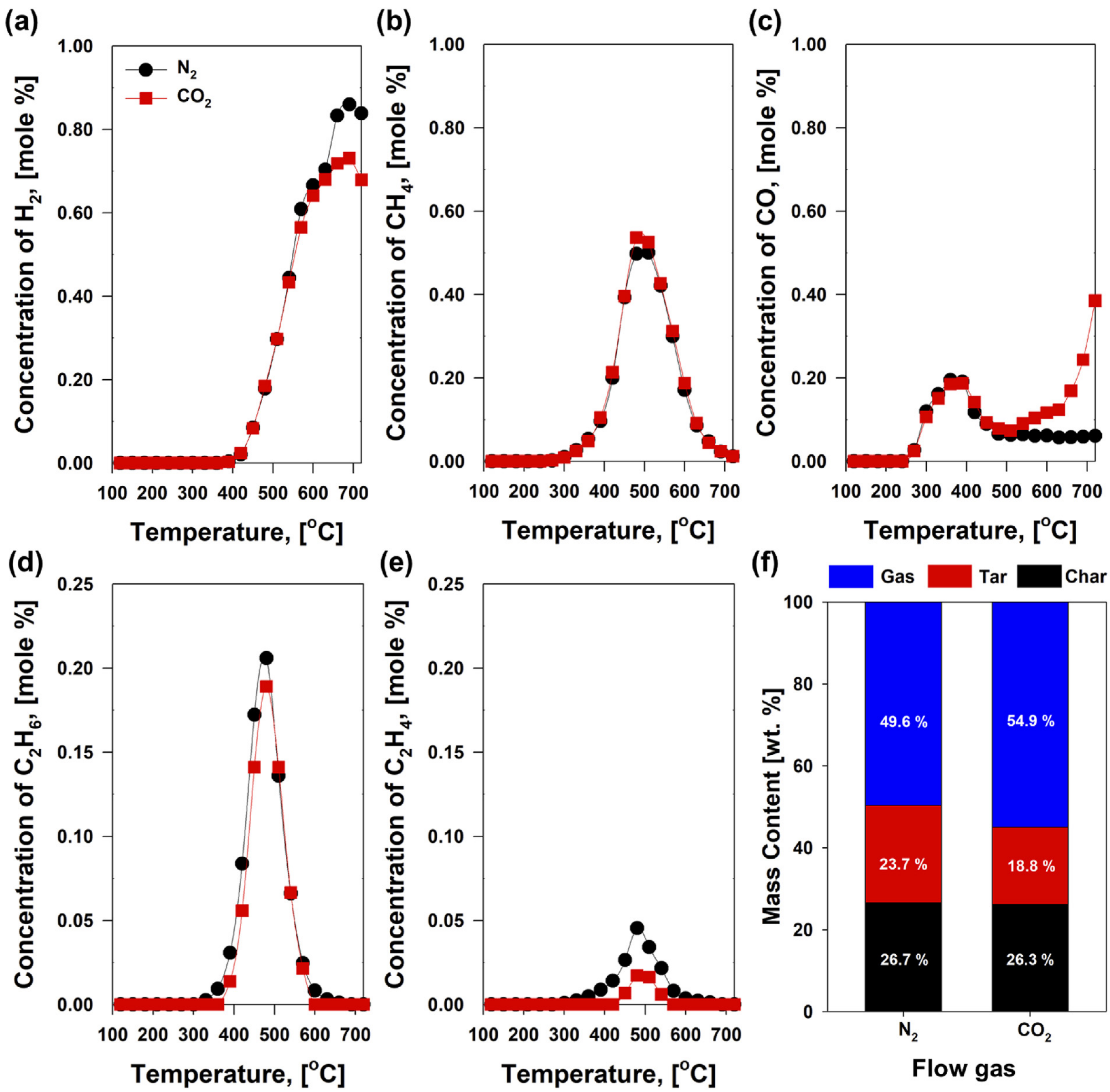

Fig. 2. (a–e) Evolution profiles of gaseous pyrolysates from *C. vulgaris* one-stage pyrolysis, and (f) overall mass balances resulted from three pyrogenic products in the N_2/CO_2 atmospheres.

biochar). To confirm this hypothesis, the mass portion of biocrude must be reduced from the CO_2 environment due to salient mechanistic feature of CO_2. Indeed, such the hypothesis is well evidenced by the overall mass balance of three pyrolysates from *C. vulgaris* pyrolysis in Fig. 2(f). As well demonstrated, the mass portions of biochars under 2 atm are nearly identical, while there is a significant difference in the mass portions of biocrudes. For instance, the mass portion of biocrude from the CO_2 environment was 18.8 wt%, while it was 23.7 wt% from the N_2 environment. As noted, the mass portion of biochar was nearly same in both environments. As such, morphologies of biochars from pyrolysis of C. *vulgaris* in both the atmospheric conditions were not different according to SEM images (Fig. 3).

3.3. *C. vulgaris pyrolysis in the two-stage reactor*

All experimental results shown in the section of 3.2 offered that the mechanistic effective of CO_2 as an oxidant was only active at $\geq$ 540 °C, and the reaction kinetics of it was accelerated as the thermolysis temperature increased. As such, the experimental setup was modified with the two-stage pyrolysis setup. For example, the first heating zone operated identically with the pyrolysis setup in the section of 3.2, and the additional heating unit attached, isothermally ran at 600 °C. The isothermal temperature (600 °C) was selected since the functional effectiveness of CO_2 was began at $\geq$ 540 °C from Fig. 2(c). This setup was intended to promote the homogeneous gas phase reaction of VMs with CO_2 at 600 °C through the second heating stage. The concentration levels of the gaseous pyrolysates originated from two-stage pyrolysis are plotted in Fig. 4(a)–(e).

In comparison with the one-stage pyrolysis results in Fig. 2(a–e), the quantity of gaseous products is indeed enhanced using two-stage setup (Fig. 4). The concentrations of H_2, CH_4, CO, and C_2H_4 more than doubled (CH_4) or had a few orders of

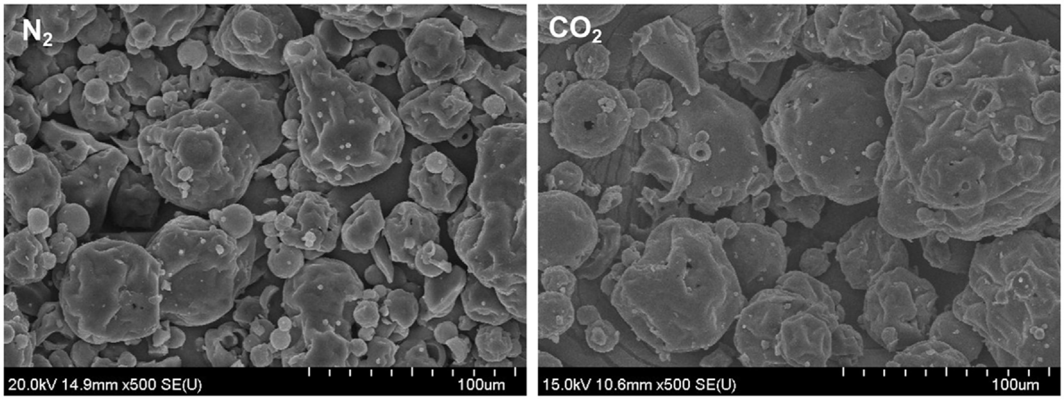

Fig. 3. SEM visualizations of biochars obtained from *C. vulgaris* pyrolysis in the N_2/CO_2 atmospheres.

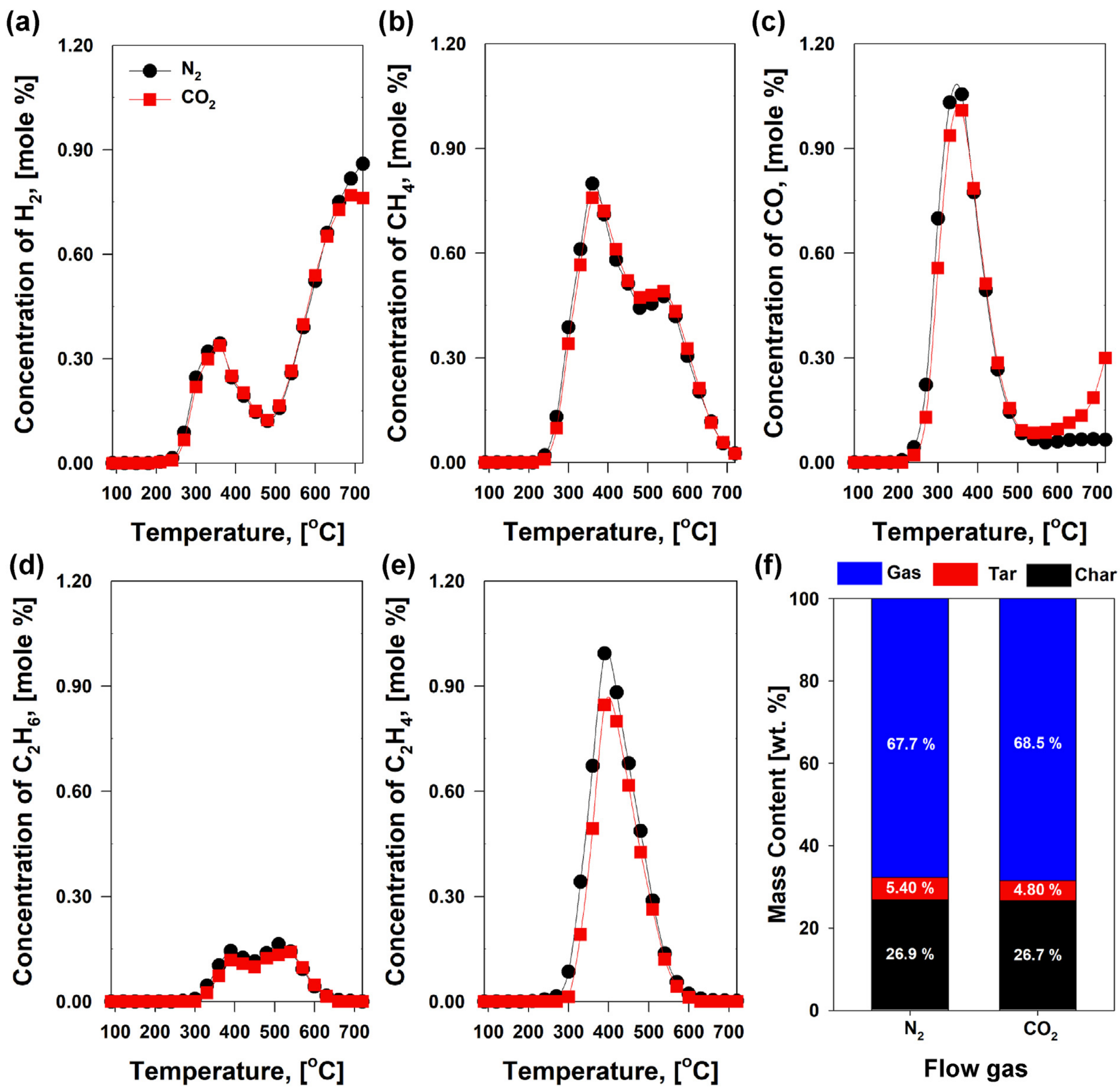

Fig. 4. (a–e) Evolution profiles of gaseous pyrolysates from *C. vulgaris* two-stage pyrolysis, and (f) overall mass balances of three pyrogenic products in the N_2/CO_2 atmospheres.

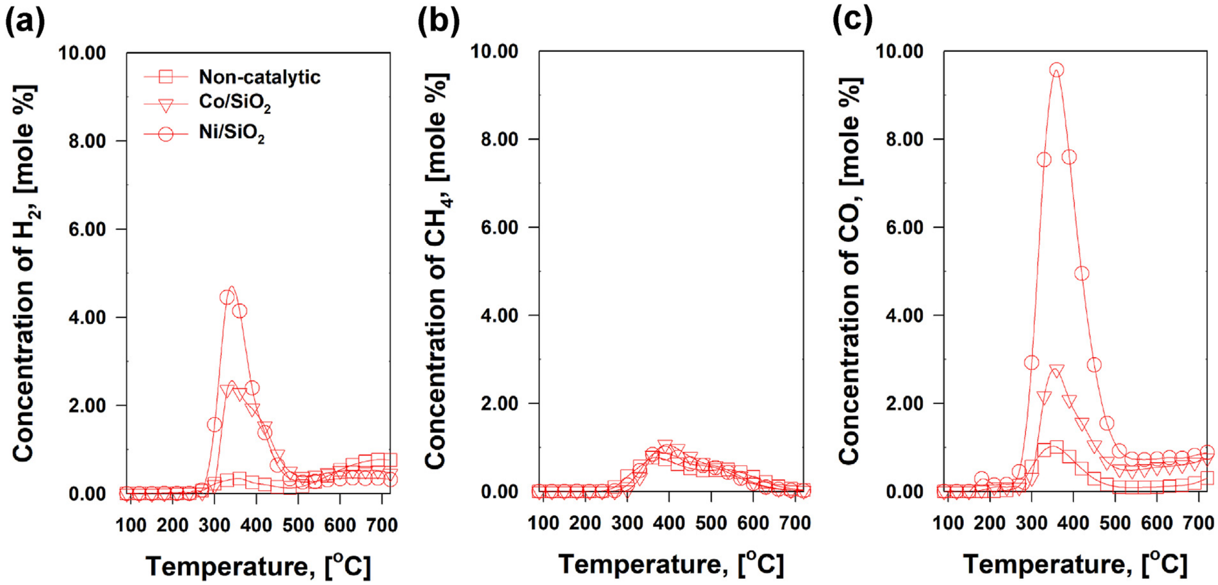

Fig. 5. (a–c) Evolution profiles of major gaseous products from catalytic pyrolysis of *C. vulgaris* with Co/SiO_2 (reverse triangle) and Ni/SiO_2 (circle), in reference to non-catalytic pyrolysis (square) under the CO_2 atmosphere.

magnitude higher (C_2H_4) in reference to one-stage reaction between 270 and 510 °C. H_2 evolution did not appear at $\leq$ 420 °C in one-stage pyrolysis, while it was initiated from 270 °C in two-stage pyrolysis. As noted in the TGA results (Fig. 1), the initiation temperature for production of gaseous products was $\geq$250 °C. Thus, it can be inferred that VMs and gaseous effluents from pyrolysis of *C. vulgaris* in the first heating unit were brought to the second unit running at 600 °C, followed by the thermal cracking of VMs to produce such the light H_2, CO, and hydrocarbons. This resulted in the significant reduction of liquid mass fraction through the two-stage pyrolysis of *C. vulgaris* than the one-stage pyrolysis (Fig. 4(f)). However, the results of gaseous pyrolysates and overall mass balances were not different from the two atmospheric conditions (Fig. 4). This indicates that reaction kinetics for the production of gaseous pyrolysates from two-stage setup were enhanced by the thermal cracking of VMs at 600 °C in the second heating stage.

3.4. Catalytic pyrolysis of C. vulgaris over cobalt/biochar composites in the CO_2 environment

The reaction kinetics for homogeneous reaction of VMs and CO_2 even at 600 °C was confirmed from *C. vulgaris* pyrolysis. This indicates that increasing the pyrolysis temperature can be an option to enhance reaction kinetics. However, higher temperature requires an additional input of thermal energy. To further expedite the reaction kinetics at 600 °C instead of increasing the pyrolysis temperature, catalytic pyrolysis was performed over 5 wt% Ni/SiO_2 and 5 wt% Co/SiO_2 under the CO_2 environment. The evolution trends of major gaseous pyrolysates from *C. vulgaris* catalytic pyrolysis under the CO_2 condition are compared in Fig. 5. C_2 hydrocarbons were not drawn because their evolution profiles were nearly identical with those from non-catalytic *C. vulgaris* pyrolysis in the two-stage setup (Fig. 4).

Evolution of CH_4 from catalytic pyrolysis of *C. vulgaris* was nearly identical that from non-catalytic pyrolysis (Fig. 5(b)). The notable difference was observed from the H_2 and CO evolution profiles (Fig. 5(a) and (c)). In comparison with non-catalytic pyrolysis of *C. vulgaris*, more than 3.1 and 2.4 times of H_2 formations was shown over both Ni/SiO_2 and Co/SiO_2 catalysts, respectively. In addition, CO formations were also substantially improved in the presence of Ni/SiO_2 (8.2 times more) and Co/SiO_2 (2.8 times) catalysts in reference to non-catalytic pyrolysis (Fig. 5(c)). Thus, it can be inferred that both catalysts expedited the dehydrogenation and deoxygenation.

4. Conclusion

Pyrolysis of *C. vulgaris* was conducted to recover energy as gaseous products under the CO_2 atmosphere. The CO_2-feeding pyrolysis substantially enhanced CO generation via gas phase homogeneous reaction with VMs at $\geq$ 520 °C. Ni/SiO_2 and Co/SiO_2 catalysts substantially improved reaction kinetics for dehydrogenation and deoxygenation, thereby resulting in the considerable improvement of syngas formations, controlling the carbon to hydrogen ratio depending on the reaction conditions.

Author contribution statement

S. Jung: Investigation, Formal analysis, Writing – original draft preparation, Data curation. J.-H. Kim: Investigation, Formal analysis, Data curation. Y.J. Jeon: Resources, Data curation, Formal analysis. Y.-K. Park: Resources, Data curation, Formal analysis. E.E. Kwon: Conceptualization, Supervision, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This study was supported by the National Research Foundation of Korea (NRF) grants funded by the Korea government (MSIT) (NRF-2019R1A4A1027795 and NRF-2019H1D3A1A01070644).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2020.118710.

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